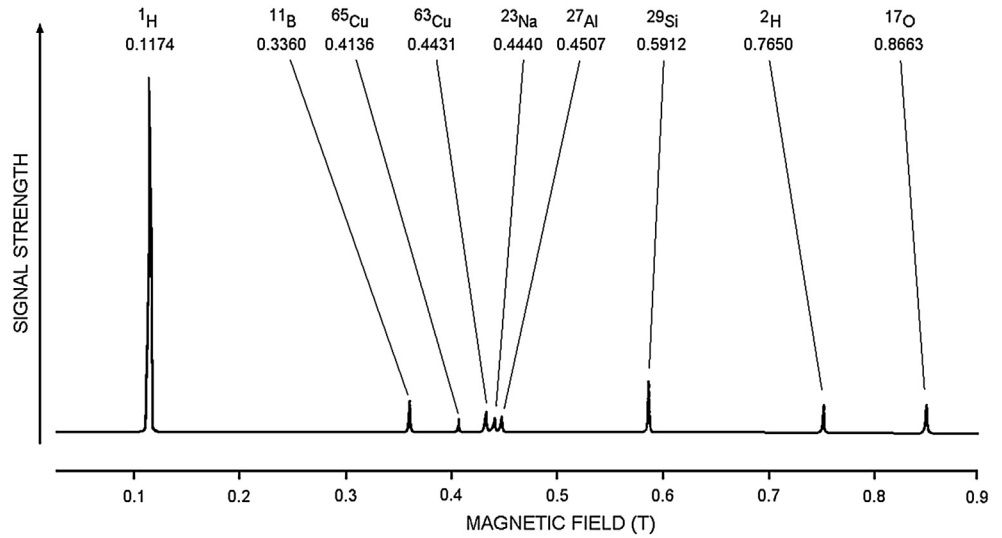


Figure 6.11 Nmr spectrum for a glass vessel containing water and some copper. The magnetic field is given in tesla (T).

nmr technique has therefore become a central tool for the investigation of chemical structures in solids, liquids, and the gaseous state. Tomography by nmr is also a valuable tool for medical imaging. To avoid the unease of the patients caused by the word "nuclear", nmr in hospitals is now usually called magnetic resonance, mr.

6.7. RADIOACTIVE DECAY AND NUCLEAR STRUCTURE

In the preceding section we have described three methods of determining nuclear spin - one optical and two magnetic. The nuclear spin plays a central role in forming the nuclear energy states. It is therefore to be expected that it also should be of importance in nuclear reactions and in radioactive decay. Let us consider some rules for the lifetimes of unstable nuclei, for their permitted modes of decay, and for the role of nuclear spin. Knowing these rules, it is, for example, possible from a decay scheme to predict the spin states of levels which have not been measured.

6.7.1. Gamma-decay

Photons are emitted in the transition of a nucleus from a higher energy state (level) to a lower

$$E_\gamma = \mathbf{h}\upsilon = E_\mathrm{f} - E_\mathrm{i} \tag{6.47}$$

where f and i refer to the final and initial states. Because the photon has spin 1, de-excitation through γ-emission is always accompanied by a spin change $\Delta I = 1$. This

Table 6.7 Classification of radiation emitted in γ-decay and lifetime calculations of the excited states for given γ-energies

Type of radiation	Name of transition	Spin change $\Delta I = \Delta l$	Parity change	Average lifetime (τ) in seconds for γ-energy level		
				1MeV	0.2 MeV	0.05MeV
E1	Electric dipole	1	Yes	4×10^{-16}	5×10^{-14}	3×10^{-12}
M1	Magnetic dipole	1	No	3×10^{-14}	4×10^{-12}	2×10^{-10}
E2	Electric quadrupole	2	No	2×10^{-11}	6×10^{-8}	6×10^{-5}
M2	Magnetic quadrupole	2	Yes	2×10^{-9}	5×10^{-6}	5×10^{-3}
E3	Electric octupole	3	Yes	2×10^{-4}	2	70 h
M3	Magnetic octupole	3	No	2×10^{-2}	180	200 d

(From Blatt and Weisskopf.)

leads to a change in the charge distribution of the nucleus and hence also to a change in its magnetic properties. Depending on the type of change occurring through the γ-emission, the radiation is classified as electric or magnetic (§6.4) according to the scheme in the left part of Table 6.7. Although parity need not be conserved in γ-decay, the parity change associated with the different types of γ-transitions is listed in that table.

Based on the single-particle model, Blatt and Weisskopf have calculated probable lifetimes for excited states assuming a model nucleus with a radius of 6 fm. For 2^L-multipole transitions of electric (E) or magnetic (M) type they derived the following equations

$$\lambda_{EL} = 4.4 \times 10^{21}\left\{(L+1)/\left[L((2L+1)!!)^2\right]\right\}\{3/(L+3)\}^2\left(E_\gamma/197\right)^{(2L+1)}r^{2L} \tag{6.48}$$

$$\lambda_{ML} = 1.9 \times 10^{21}\left\{(L+1)/\left[L((2L+1)!!)^2\right]\right\}\{3/(L+3)\}^2\left(E_\gamma/197\right)^{(2L+1)}r^{2L-2} \tag{6.49}$$

where L is the angular momentum carried away by the photon ($= \Delta l$), λ is the decay constant in s^{-1}, E_γ is the γ-ray energy in MeV and r is the nuclear radius in fm.

As seen, for decay involving electric multipole transitions the average lifetime τ is proportional to $r^{-2\Delta l}$, and for decay involving magnetic multipoles, to $r^{-2(\Delta l-1)}$; in the decay the nuclear spin quantum number s does not change. Calculated values are included in Table 6.7.

The decay of ^{24m}Na ($I = 1+$) to ^{24}Na ($I = 4+$) provides a useful example. The transition involves $\Delta l = 3$, no (i.e. there is no parity change), so it is designated as M3. The γ-energy is 0.473 MeV and the nuclear radius of ^{24}Na is about 3.7 fm. In order to compare this energy with those given in Table 6.7, a radius correction from the assumed 6 fm to the observed 3.7 fm must be made; accordingly $E_{corr} = E_{obs}(r_i/6)^{-2(\Delta l-1)}$ for

M-type radiation. This gives a hypothetical energy of $0.473(3.7/6)^{-4} = 3$ MeV. According to Table 6.7 the lifetime should be >0.02 s; the observed value is 0.035 s. The agreement between experiment and calculation is sometimes no better than a factor of 100.

The Blatt-Weisskopf relationship between energy and lifetime is only applicable for excited nucleonic states, not for rotational states. In the decay of these states the rotational quantum number always changes by two units and the lifetime of the states is proportional to $E_\gamma^{-5}\ \hat{Q}^{-2}$. The lifetimes are so short that no rotationally excited isomers have been observed as yet.

6.7.2. Beta-decay

Beta-decay theory is quite complicated and involves the weak nuclear interaction force, which is less understood than the strong interaction. The theory for β-decay derived by Fermi in 1934 leads to the expression

$$\lambda = G|M|^2 f \tag{6.50}$$

for the decay constant, λ. G is a constant, $|M|$ is the nuclear matrix element describing the change in the wave function during the β-transformation (i.e. of a proton into a neutron for positron emission, or the reverse for negatron emission), f is a function of E_{max} and Z. $|M|$ depends on the wave functions before and after the transformation and gives the "order" of decay. Since $t_{½} = \ln 2/\lambda$

$$\ln 2 = G|M|^2 f t_{½} \tag{6.51}$$

we see that the product $ft_{½}$ should be constant for a decay related to a certain $|M|$. The ft value (omitting index ½) is often referred to as the comparative β half-life, and nomograms for its calculation are given in nuclear data tables and decay schemes. The lower the ft value the higher is the probability for decay, and the shorter is the half-life. Gamow and Teller have given selection rules for β-decay which are useful for estimating decay energy, half-life, or spin in a certain decay process, if two of these properties are known. These rules are summarized in Table 6.8.

Table 6.8 Gamow Teller Selection Rules for β-Decay

Transition type	Δ*I*	Parity change	Log *ft*
Super allowed	0	No	3
Allowed	0, ±1[(a)]	No	4-6
First forbidden	0, ±1, ±2	Yes	6-9
Second forbidden	±2, ±3[(b)]	No	10-13

[(a)] Not $0 \rightarrow 0$
[(b)] Also $0 \rightarrow 0$.

For example, the decay of ^{24}Na occurs 99% through β-emission (with an $E_{max} = 1.4$ MeV) to an excited state of ^{24}Mg (Fig. 4.7). The log *ft* value of the transition is 6.1. The ground state of ^{24}Mg is 0+; the excited state has positive parity. Thus the selection rules indicate an allowed transition for which the only spin changes permitted are 0 and ±1. The ground state of ^{24}Na is 4+, and the observed excited state (at the 4.12 MeV level) is 4+, in agreement with the rule.

6.7.3. Alpha-decay theory

If we calculate the *Q*-value for the α-decay reaction (5.13) from the mass formulae, (4.15), we find that $Q > 0$ for all nuclei with $A > 150$, which means that we would expect all elements heavier than the rare earths to be unstable with respect to α-decay. However, the accuracy of (4.15) decreases when we move away from the region of β stability. Hence, we should not be surprised that nuclides, with a high Z/A ratio, far from stability have been found to decay by α-emission down to $A = 106$. For nuclei nearer β-stability, decay by emission of α-particles is observed for some isotopes of rare earths and heavier elements, but it occurs frequently only for $A = 210$, i.e. nuclides heavier than ^{209}Bi.

In §4.17 we mentioned the discovery by Geiger and Nuttall that the lower the α-energy the longer was the half-life of the α-decay; doubling the decay energy may reduce the half-life by a factor of 10^{20}. Alpha-decay has been observed with energies from slightly greater than 1.8 MeV (e.g. ^{144}Nd, $E_\alpha = 1.83$ MeV, $t_{½} = 2.1 \times 10^{15}$ y) to about 10 MeV (e.g. ^{262}Ns, $E_\alpha = 10.38$ MeV, $t_{½} = 4.7$ ms). The isotopes of the actinide elements typically have α-energies between 4 and 10 MeV.

The observed stability against α-decay for most nuclei in the range $102 \leq A \leq 210$ having a positive Q_α can be explained by assuming the α-particle exists as a (preformed) entity inside the nucleus but with insufficient kinetic energy to overcome the "internal Coulomb barrier". This barrier is assumed to be of the same type, although of somewhat different shape, as the external Coulomb barrier, which is discussed in some detail in §10.3.

Assume that the average kinetic energy is at the level marked E_α in Figure 6.12. If the particles in the nucleus have a Boltzmann energy distribution, an α-particle could in

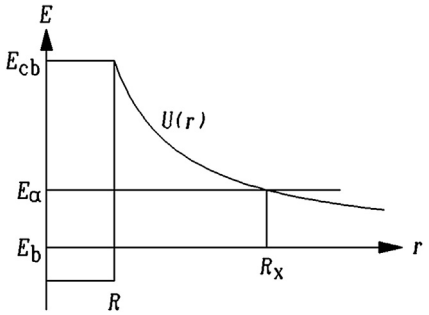


Figure 6.12 Alpha-penetration through the potential wall.

principle form in the nucleus and perhaps acquire sufficient kinetic energy through collisions to overcome the barrier (E_{cb}). It would then be emitted with an energy E_{cb}, which for an element like uranium is 26 MeV. However, the observed α-energy is only 4.2 MeV. This reasoning is very similar to chemical reactions having to pass an activation energy to proceed and that reaction component energy follow a Boltzmann distribution in energies. Thus, a chemical reaction rate can be determined from the activation energy and the half life of an alfa decay can be estimated from the alpha energy, i.e. essentially from the value of the α-particle wave function outside the coulomb barrier.

This contradiction was explained by Gamow, and independently by Gurney and Condon, in 1928, by using a quantum mechanical model, which retained the feature of the "one-body model" with a preformed α-particle inside the nuclear potential well of even-even nuclei.

The time independent solution to the Schrödinger wave equation for an α-particle inside the nuclear potential well is a wave function which has a small, but non-zero, value even outside the potential well. The probability, p, to find the α-particle outside the potential well is the square of the wave function and is found to be

$$p = \exp\left[-\left\{4\pi\sqrt{2\mu}/\mathbf{h}\right\} \int_{R}^{R_x} \sqrt{U(r) - Q_\alpha}\,\mathrm{d}r \right] \tag{6.52}$$

where index α refer to the α-particle, μ is the reduced mass of α-particle and residual nucleus, $\mu = M_\alpha\, M_1/(M_\alpha + M_1)$, R and R_x are inner and outer integration limits respectively (where the potential energy of the barrier is equal to the energy of the emitted particle, cf. Fig 6.12), r is the distance from the center of the nucleus, $U(r)$ is the potential energy of the α-particle, and Q_α is the total α-decay energy. Index 1 refer to the nucleus remaining after emission of the α-particle.

The decay constant can be regarded as the product of p and the frequency, f, by which the α-particle hits the barrier from inside. If we assume that the deBroglie wavelength, $\mathbf{h}/\mu v$, for an α-particle of velocity v inside the nucleus is approximately equal to the nuclear radius, R, we obtain

$$\mathbf{h}/\mu v \approx R \tag{6.53}$$

The frequency f can then be estimated from v if we assume that the α-particle bounces back and forth inside the potential well with constant velocity.

$$f = v/2R \approx \mathbf{h}/2\mu R^2 \tag{6.54}$$

Combining (6.40) with (6.38) we obtain the following expression for the decay constant

$$\lambda \approx \left[\mathbf{h}/2\mu R^2\right] \exp\left\{ -\left[4\pi\sqrt{2\mu}/\mathbf{h}\right] \int_{R}^{R_x} \sqrt{U(r) - Q_\alpha}\,\mathrm{d}r \right\} \tag{6.55}$$

The same relation also holds – with appropriate substitutions – for any charged particle trying to enter the nucleus from outside the potential barrier (cf. §10.3), where, however, only one impact is possible, i.e. $f = 1$.

For some simple mathematical forms of the potential energy $U(r)$ it is possible to find analytical solutions to the integral in (6.55). The simplest form of $U(r)$, a square well nuclear potential according to Figure 6.12, yields the following expression after integration and some algebra[5]

$$\lambda \approx \left[\mathbf{h}/\left(2\mu R^2\right)\right] \exp\left\{ - \left[\sqrt{2\mu}\mathrm{e}^2 Z_1 Z_\alpha / \left(\varepsilon_0 \mathbf{h}\sqrt{Q_\alpha}\right)\right] \left[\mathrm{asccos(u)} - \mathrm{u}\sqrt{1-u^2}\right]\right\} \tag{6.56}$$

where

$$u = \left(E_\alpha / E_{\mathrm{cb}}\right)^{1/2} = \left[4\pi\varepsilon_0 Q_\alpha R / \left(Z_1\ Z_\alpha\ \mathbf{e}^2\right)\right]^{1/2} \tag{6.57}$$

The radius of the decaying nucleus can be estimated from

$$R = r_0 A_1^{1/3} + r_\alpha \approx 1.30\, A_1^{1/3} + 1.20 \ \ (\mathrm{fm}) \tag{6.58}$$

where A_1 is the mass number of the nucleus after emission of the α-particle and r_α is the effective radius of the α-particle. Equation (6.56) can not only be used to compute decay constants, but also to estimate the nuclear radius for even-even α-emitters from measured half-lives and decay energies. The calculated decay constant is very sensitive to Q_α: a 1 MeV increase in Q_α increases λ (and decreases $t_{½}$) by a factor of about 10^5. It is also very sensitive to the nuclear radius: a 10% increase in R (or the Coulomb radius r_c; see Fig. 12.4) which means a corresponding decrease of the Coulomb barrier height, increases λ by a factor of 150.

Decay constants and partial α half-lives computed from (6.56) are normally within a factor 4 of the measured values for even-even nuclei. The half-lives of even-odd, odd-even, and odd-odd nuclides are often longer than predicted by equations like (6.56), even after inclusion of more elaborate nuclear potentials and angular momentum effects in the theory. The ratio between the observed and predicted half-lives is called *hindrance factor* and ranges from one to ~3000.

Assuming the Coulomb barrier to be much larger than Q_α, i.e. u small, designating the resulting exponential term in (6.56) by e^{-2G} and taking the logarithm of the resulting expression we obtain

$$\log \lambda = \mathrm{constant} - 2G \tag{6.59}$$

which is very similar to the empirical Geiger-Nuttall law.

[5] It is important to recognize the difference between the classical CGS system where the force F between point charges in vacuum is $F = z_1\ z_2\ \mathbf{e}^2/r^2$ and the SI system where $F = z_1\ z_2\ \mathbf{e}^2/(4\pi\varepsilon_0\ r^2)$.

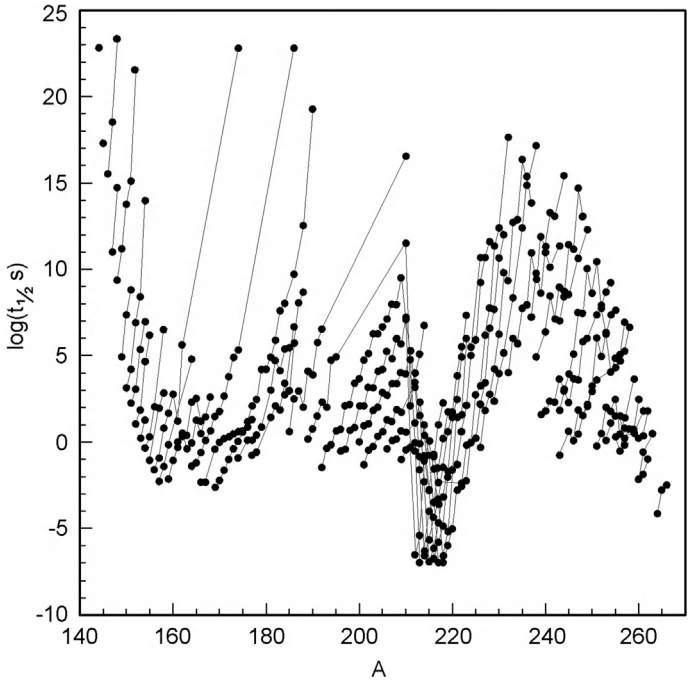


Figure 6.13 The variation of α-half-life with mass number. Lines connect isotopes.

Figure 6.13 shows the systematic change of α-decay half-life with nuclear charge and mass for heavy nuclei, lines connect data for constant Z. Odd-even effects and influence from magic numbers are visible. Such diagrams have historically played an important role in the synthesis and identification of isotopes of the heaviest elements.

The α-decay theory was the first successful (quantum mechanical) explanation of radioactive decay, and as such played a major role in further development of nuclear theories and models. Although its simplicity causes it to fail for non-spherical nuclei as well as those near closed shells, such effects can be taken into account in more advanced Nilsson-type calculations.

6.7.4. Spontaneous fission

In §5.4 we found that fission of heavy nuclei like ^{236}U is exoergic and in §5.6 that it is a common decay mode for the heaviest nuclei. From the semiempirical mass equation (4.15) it can be found that fission of nuclei with $A \geq 100$ have positive Q-values. Why is the decay by spontaneous fission only observed for nuclei with $A \geq 230$?

The breakup of a large even-even nucleus into two positively charged fragments of roughly equal mass and charge can be treated in a way similar to that of α-decay.

Assuming separation into two spherical fragments, $A_1\ Z_1$ and $A_2\ Z_2$, in point contact the Coulomb energy can be calculated to be

$$E_{cb} = 0.96\ Z_1\ Z_2 / \left(A_1^{1/3} + A_2^{1/3} \right) \quad \text{(MeV)} \tag{6.60}$$

where $r_0 = 1.5$ fm has been assumed. In a real case it is necessary to consider that the newly formed fragments have non-spherical shapes and the value obtained from (6.46) is thus very approximate but sufficient for our discussion. The spontaneous fission of a nucleus is obviously hindered by a Coulomb barrier, the *fission barrier*, and the process should be treated as a barrier penetration problem for $Q < E_{cb}$. When $Q = E_{cb}$, breakup of the nucleus will occur within a few nuclear vibrations, $\sim 10^{-22}$ s.

The critical condition $Q = E_{cb}$ for fission of an even-even nucleus into two equal fragments with an unchanged charge to mass ratio can be estimated by equating E_{cb} from (6.46) with the Q-value computed from (3.8). Neglecting the pairing term this results in

$$\left(Z^2/A \right)_{crit} = 37.89 \tag{6.61}$$

Because asymmetric fission is more common than symmetric and the emerging fission fragments have non-spherical form, the numerical value derived above is not very accurate. However, the concept of a critical value of Z^2/A is important. A more sophisticated treatment results in the equation

$$\left(Z^2/A \right)_{crit} = 50.883 \left[1 - 1.7826\ (N - Z)^2 / A^2 \right] \tag{6.62}$$

We can then define a *fissionability parameter*, x,

$$x = \left(Z^2/A \right) / \left(Z^2/A \right)_{crit} \tag{6.63}$$

as a measure of how prone to fission a nucleus is.

Figure 6.14 shows fission barrier height as function of Z^2/A. Calculations by Myers and Swiatecki using a refined liquid drop model predict that the barrier height should pass through a maximum around $Z^2/A \sim 16$. The penetrability of the barrier increases roughly exponentially with its height. Hence, low values of Z^2/A (but above the value corresponding to the maximum barrier height) implies extremely long half-lives. As an example, ^{238}U has $Z^2/A = 35.56$, $E_{cb} = 5.8$ and a partial half-life for spontaneous fission, $t_{½,SF}$, of $\sim 10^{16}$ y. By comparison we can estimate that a nucleus with $A = 100$ and $Z = 44$ (^{100}Ru), $Z^2/A = 19.36$, has a fission barrier tens of MeV high and a practically infinite half-life with regard to spontaneous fission.

So far we have neglected the effects of pairing, nuclear shell structure and nuclear deformation on the fission process. Odd-even, even-odd and odd-odd nuclei exhibit

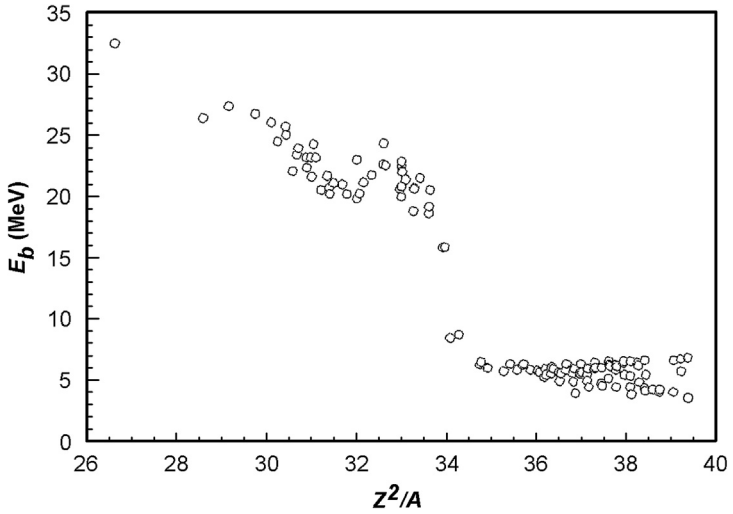


Figure 6.14 Fission barrier height as function of Z^2/A.

large hindrance factors, HF, for spontaneous fission somewhat similar to the phenomenon observed in α-decay. The presence of one or two odd nucleons leads to spin and parity values which must be conserved during the deformations leading to fission and thus constrains the possible shapes and energy levels. This contributes to the occurrence

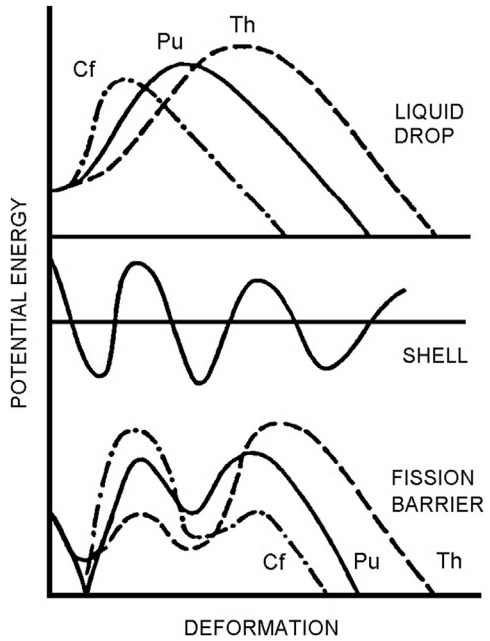


Figure 6.15 Qualitative features of the fission barrier for Pu. *(From Britt.)*

of hindrance factors. Hence, o-e, e-o and o-o nuclei have normally much longer spontaneous fission half-lives than their neighboring e-e nuclei.

The ground state energy, E, of a nucleus can be regarded as a sum of the liquid drop model energy (including deformation), E_{LDM}, the pairing correction, δ_{P}, and the shell correction, δ_{S}, to that energy.

$$E = E_{\mathrm{LDM}} + \delta_{\mathrm{P}} + \delta_{\mathrm{S}} \tag{6.64}$$

The use of (6.64), with deformation dependent shell corrections, leads to fission barriers with two maxima, see Figure 6.15. The occurrence of a secondary minimum is consistent with the observation of spontaneous fission isomers. Shell and pairing effects also give rise to long spontaneous fission and α-decay half-lives for nuclides around magic N or Z numbers. Detailed calculations based on advanced theories of nuclear structure lead to the prediction of an area of spheroidal nuclei with increased nuclear stability (in their ground state) around $Z = 114$ and $N = 184$, the so called *superheavy elements*, see Ch. 14.

A number of very rare decay modes have also been observed where clusters of nucleons heavier than α-particles are emitted, see §5.7. Some of these are difficult to distinguish from highly asymmetrical fission.

6.8. EXERCISES

6.1. The quantum numbers $s = ½$ and $l = 2$ are assigned to a particle. (a) If spin and orbital movements are independent, how many space orientations (and thus measured spectral lines if no degeneration of energy states occur) are possible in an external field of such a strength that both movements are affected? (b) How many lines would be observed if spin and orbital movements are coupled?

6.2. In the hydrogen atom the K-electron radius is assumed to be 0.529×10^{-10} m (the Bohr radius). (a) Calculate the orbital velocity of the electron assuming its mass to be $\mathbf{m}_{\mathrm{e}}$. (b) How much larger is its real mass because of the velocity? Does this affect the calculations in (a)?

6.3. A beam of protons pass through a homogeneous magnetic field of 0.5 T. In the beam there is a small high frequency coil which can act on the main field so that the proton spin flips into the opposite direction. At which frequency would this occur?

6.4. Calculate the nuclear Landé factor for ^{11}B.

6.5. In Table 6.5 the first degenerate levels have been given. Using the same assumptions, what states will be contained in the next level and how many nucleons will it contain?

6.6. How deep is the nuclear well for ^{116}Sn if the binding energy of the last nucleon is 9 MeV?

6.7. Calculate the spins and nuclear g factors for (a) ^{45}Ca, (b) ^{60}Co, and (c) ^{141}Pr, using data in Table 6.3.